M.Sc. Cyber Security Lab Assignment – LAB 2

Course: Cyber Security / Information Security
Class: M.Sc. Computer Science / Cyber Security
Lab Duration: 2 Hours
Lab No.: 2
Lab Title: Password Security, Malware Awareness and Risk Analysis

Objective

To understand password vulnerabilities, password attacks, malware types, and risk assessment using practical tools and case analysis.

Instructions to Students

1. Perform all tasks practically.
2. Paste clear screenshots for every task.
3. Use Kali Linux where required.
4. Do not perform attacks on external or unauthorized systems.
5. Submit the completed lab file after execution.

Task 1: Password Strength Analysis

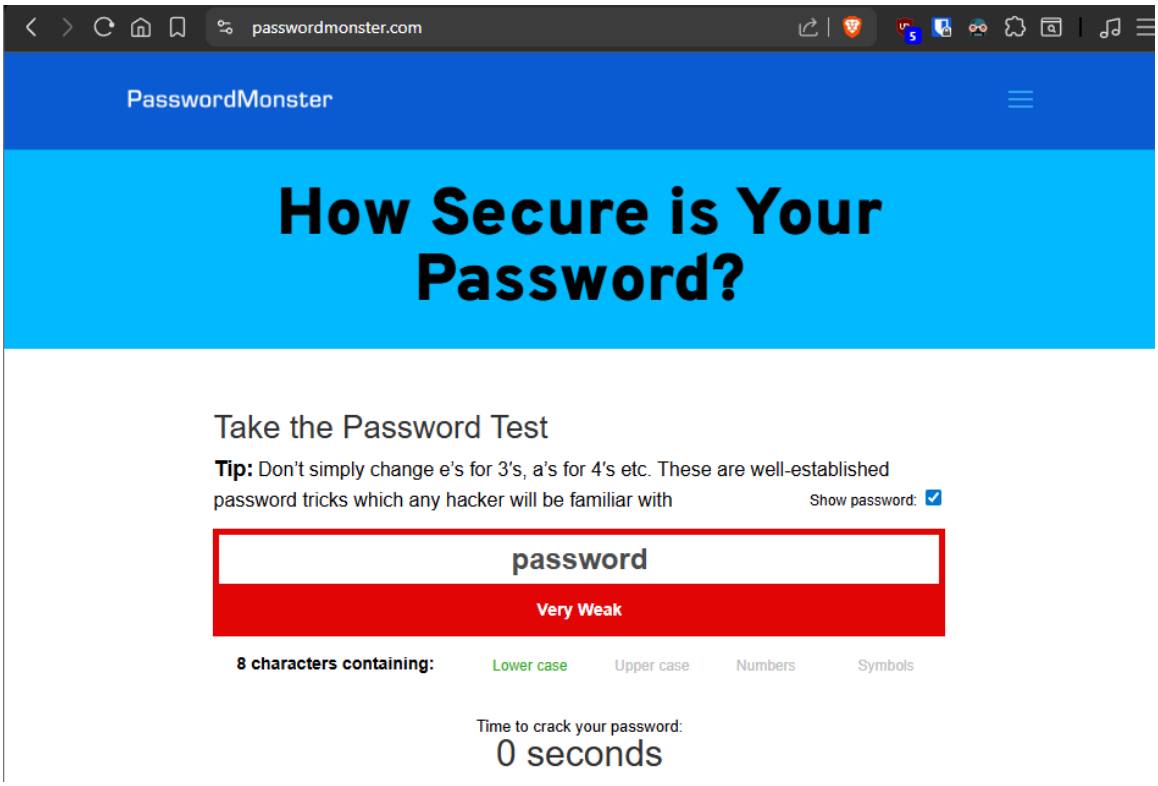
Tool Used: Online Password Strength Checker
Website: https://www.passwordmonster.com

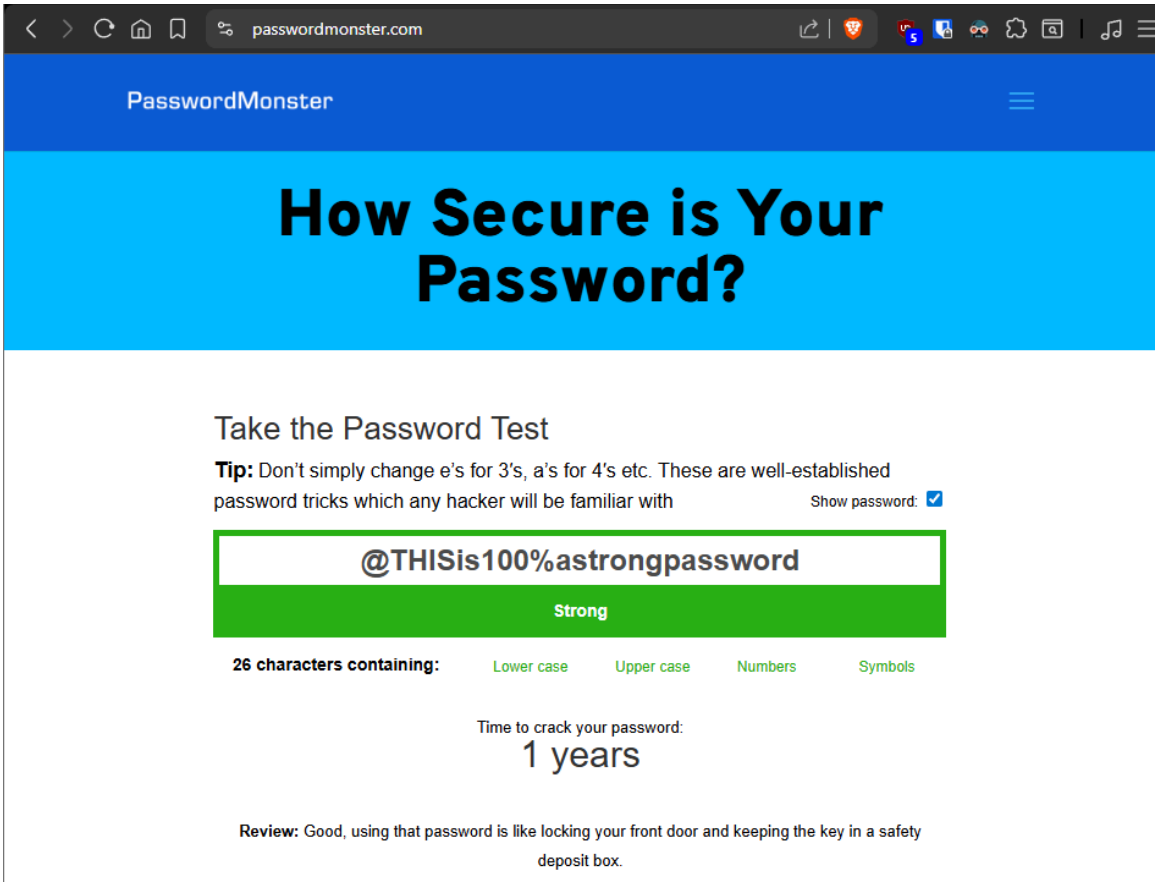
Steps:
a) Enter a weak password: **password**
b) Enter a strong password: **@THISis100%astrongpassword**
c) Compare cracking time.

Weak password: **0 seconds**

Strong Password: **1 year**

d) Paste screenshots.


passwordmonster.com
PasswordMonster
How Secure is Your Password?
Take the Password Test
Tip: Don't simply change e's for 3's, a's for 4's etc. These are well-established password tricks which any hacker will be familiar with
Show password:
password
Very Weak
8 characters containing:
Lower case
Upper case
Numbers
Symbols
Time to crack your password:
0 seconds



Observation:

1. The first password(password) was classified as very weak. It only contained lower case letters. It was a very predictable and common word.
2. The second password(@THISis100%astrongpassword) was classified as strong. It has 26 letters which significantly increases security. It also used a mix of uppercase, lowercase letters, numbers and symbols.
3. This highlights that short, simple passwords are extremely insecure. Long passwords with mixed character types are far more resilient to cracking attempts.

Task 2: Password List Generation using Crunch

Tool Used: Crunch (Kali Linux)

Command Executed:
crunch 4 6 abcd12 -o passwords.txt

```
kali@kali: ~
Session  Actions  Edit  View  Help
┌──(kali㉿kali)-[~]
└─$ crunch 4 6 abcd12 -o passwords.txt
Crunch will now generate the following amount of data: 379728 bytes
0 MB
0 GB
0 TB
0 PB
Crunch will now generate the following number of lines: 55728

crunch: 100% completed generating output
```

```
*~/passwords.txt - Mousepad
File  Edit  Search  View  Document  Help
1 Saaaa
2 aaab
3 aaac
4 aaad
5 aaa1
6 aaa2
7 aaba
8 aabb
9 aabc
10 aabd
11 aab1
12 aab2
13 aaca
14 aacb
15 aacc
```

```
55723 22222a
55724 22222b
55725 22222c
55726 22222d
55727 222221
55728 222222
55729
```

Observation:

1. The **crunch** tool generates a custom list of possible passwords. The specific command(*crunch 4 6 abcd12 -o passwords.txt*) generates passwords with minimum

length of 4, maximum length of 6, and characters limited to a, b, c, d, 1, 2. The generated passwords were stored in the file *passwords.txt*.

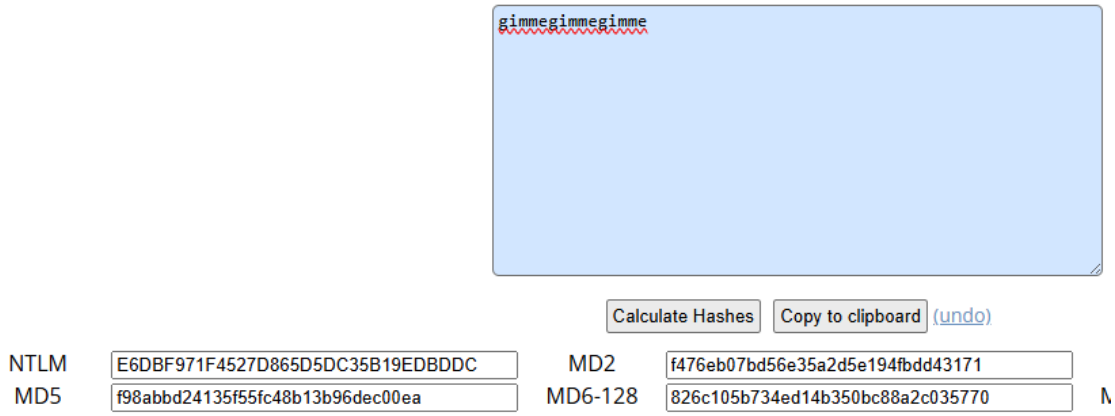
2. The output file contains 55729 generated passwords. The contents of the file show systematically generated passwords(example: aaaa, aaab, aaac, etc).

Task 3: Hash Identification

Website: https://hashes.com/en/tools/hash_identifier

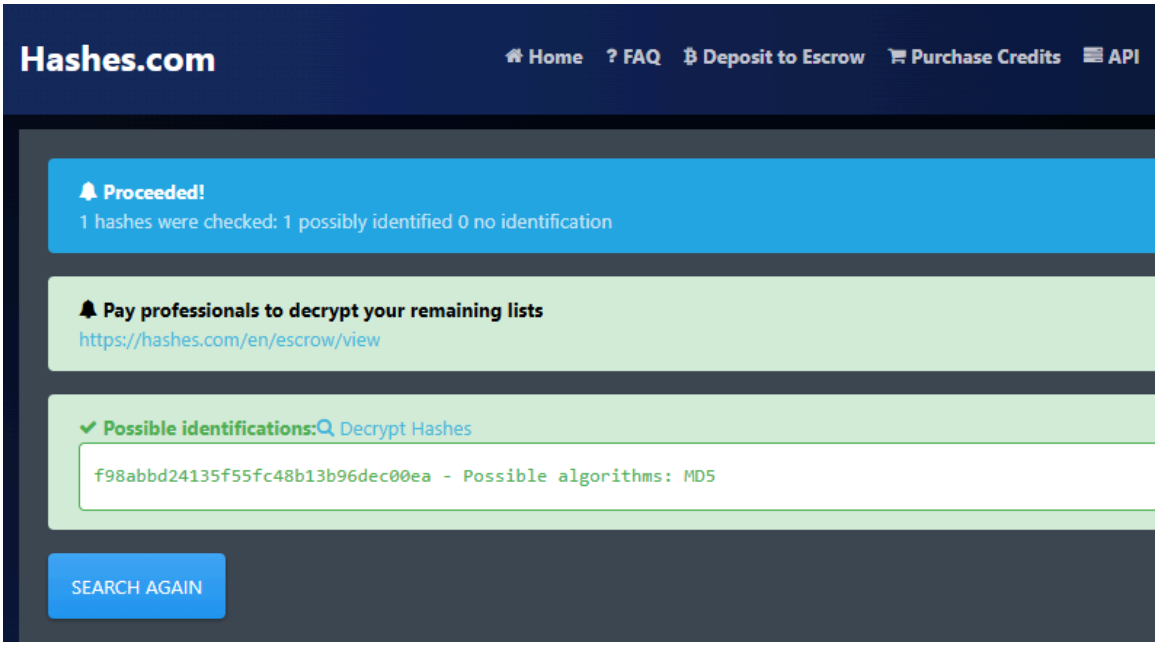
Hash Generated:

Tool used: https://www.browserling.com/tools/all-hashes



F98abbd24135f55fc48b13b96dec00ea (MD5)

Identified Hash Type:
MD5 Hash



Risk:

The tool was easily able to identify that the hash is likely MD5, once the algorithm is known, attackers can directly apply attacks optimized for it. Example: MD5 produces the same hash for the same input every time, attackers can use precomputed tables(Rainbow tables) to reverse common passwords without guessing.

Task 4: Malware Analysis (Case Study)

Selected Malware: Rootkit

Description:
A rootkit is a type of malware that hides itself and other malicious programs on a system. It allows attackers to gain unauthorized access and control while remaining difficult to detect.

Infection Method:
Rootkits usually enter a system through malicious software downloads, infected email attachments, or by exploiting security vulnerabilities in the operating system or applications.

Prevention:
Rootkit infections can be prevented by keeping the operating system and software up to date, using trusted antivirus and anti-rootkit tools, and avoiding downloads from untrusted sources.

Task 5: Risk Analysis Table

Threat	Vulnerability	Risk
Phishing	Lack of user awareness	Credential theft
Weak Password	No password policy	Account compromise
Malware	Downloading untrusted software	System infection
Rootkits	Missing security patches	Persistent unauthorized access

Result & Conclusion

This lab helped in understanding how weak passwords, poor security practices, and malware can create serious security risks. The first experiment showed that simple passwords can be cracked very quickly, while longer passwords with mixed characters are much safer. Tools like Crunch showed how attackers generate password lists, and hash identification showed why weak algorithms like MD5 are insecure. The rootkit case study explained how malware can hide itself and maintain unauthorized access to a system.

Submission Checklist

✓ Screenshots attached
✓ Observations written
✓ Commands executed
✓ File named properly